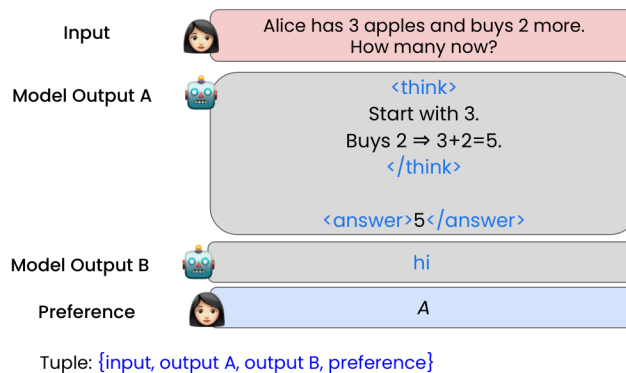


```
# ChatML Format (GPT-4, Qwen):
<|im_start|>system\nYou are helpful.<|im_end|>
<|im_start|>user\nWhat is 2+2?<|im_end|>
<|im_start|>assistant\n4<|im_end|>

# Llama 3 Format:
<|begin_of_text|><|start_header_id|>system<|end_header_id|>
You are helpful.<|eot_id|><|start_header_id|>user<|end_header_id|>
What is 2+2?<|eot_id|><|start_header_id|>assistant<|end_header_id|>
4<|eot_id|>

# In HuggingFace, tokenizer.apply_chat_template() handles this automatically!
```

Sample reward model data for RL



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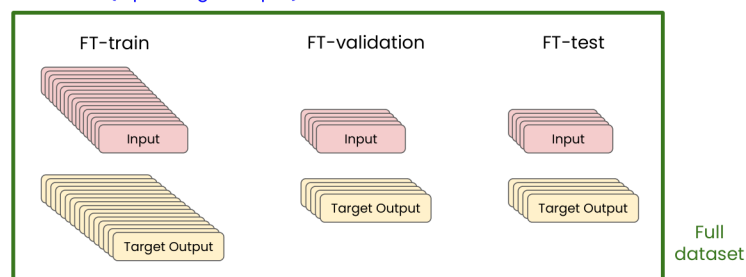
AMD

Figure 6.3: High-dimensional embedding matrix projection from discrete tokens to latent space vectors.

Sampling Dynamics: Temperature, Top-p & Top-k

Split your data to trust your LLM

- Fine-tuning data
- Train: FT-train {input, target output}
 - Hyperparameter tune: FT-validation {input, target output}
 - Eval: FT-test {input, target output}



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Figure 6.4: Logit distribution transformation via Softmax and temperature scaling.